

Ronit Shahu

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AI SCIENTIST | MACHINE LEARNING ENGINEER

PROFESSIONAL SUMMARY

AI Scientist with hands-on experience designing, training, and deploying ML, NLP, and GenAI models across the full research-to-production lifecycle. Skilled in Python, PyTorch, TensorFlow, Azure ML, and modern MLOps practices. Strong foundation in experiment design, statistical analysis, model evaluation, and scalable deployment using FastAPI, Docker, and CI/CD. Experienced in LLM integration, RAG pipelines, and advanced modelling techniques to solve complex real-world business problems. Adept at working with structured and unstructured data, collaborating with cross-functional partners, translating ML insights to stakeholders across risk, operations, and decision-making workflows and delivering measurable business impact.

TECHNICAL SKILLS, MLOps & PLATFORMS

Python · NumPy · Pandas · Scikit-learn · PyTorch · TensorFlow · Docker · Kubernetes · Azure ML · AWS · MLflow · Databricks · FastAPI · CI/CD (GitHub Actions, Jenkins) · Git · GitHub Actions · Spark · REST APIs · Cloud Monitoring · Vector Databases (FAISS, Pinecone)

ML / AI

Supervised, Unsupervised & Reinforcement Learning · NLP/NLU · Transformers · GenAI · LLM Integration · LangChain · LangGraph · RAG Pipelines · Embeddings · Vector Databases · Experiment Design · Hypothesis Testing · Statistical Modeling · Model Selection · Feature Engineering (SQL, PySpark) · Data Cleaning & Wrangling

CORE COMPETENCIES

AI-Driven Process Transformation: Improved business operations by streamlining manual work, reducing review time by 40%, improving data accuracy by 35%, and speeding up decision-making across healthcare and financial processes.

AI/ML Research Lifecycle & Leadership: Advanced from contributor to team lead, directing cross-functional teams to deliver consistent, high-quality solutions.

Continuous Improvement & Collaboration: Recognized as a top performer and innovation finalist for creating impactful solutions that improved document processing and information accuracy.

WORK EXPERIENCE

Lead Machine Learning Researcher, Canadian Robotics & AI Lab, University of Ottawa, ON | Jan 2025 – May 2025

Appointed to lead a cross-functional team contributing to national AI ethics research.

- Led a 15-member data science team through the full research lifecycle—problem framing, data engineering, model experimentation, evaluation, and production deployment—using Python, PyTorch, TensorFlow, and scikit-learn.
- Built and deployed containerized ML APIs with FastAPI, Docker, and GitHub Actions on AWS EC2 and reproducible on Azure ML, integrating MLflow tracking, CloudWatch monitoring, and version-controlled model delivery for reproducible research.
- Delivered scalable microservices with automated scheduling and monitoring, cutting compute costs by 50%. Developed transparent model evaluation reports to support scientific validation and cross-team decision-making.

Tech Stack Used: *Python, PyTorch, scikit-learn, FastAPI, Docker, GitHub Actions, AWS EC2, Azure ML, MLflow*

AI Consultant (Freelance) | Oct 2023 – Present

Provide end-to-end AI consulting to design, deploy, and optimize practical, responsible machine learning systems.

- Delivered end-to-end AI/ML consulting for small-business clients focused on enterprise automation workflows, helping them design, build, and operationalize ML systems that improved decision-making efficiency by 30–40% through responsible, production-ready solutions.
- Built and deployed ML, NLP, and GenAI models using Python, scikit-learn, TensorFlow, and PyTorch, developing

modular non-notebook pipelines for feature engineering, evaluation, and inference—boosting model accuracy by up to 25% and reducing manual review cycles by 40% across automation processes.

- Designed and containerized ML APIs using FastAPI and Docker, implementing CI/CD and AWS-based deployment workflows that reduced release times by 50%, lowered inference latency by 20–30%, and delivered scalable batch/real-time model serving to support automated business operations and ROI gains.

Tech Stack Used: *Python, scikit-learn, LangChain, FastAPI, Docker, AWS (EC2, Lambda, S3), CI/CD, Git, SQL, PySpark*

Software Developer – Life science and Health Care, Persistent Systems Limited, India | June 2022 – July 2023

- Medical Data Extraction for Clinical Trials: Designed and trained custom spaCy NER models to extract cancer-related entities, integrated with Azure ML endpoints for deployment, reducing data preparation time by 40% and improving insight accuracy.
- Healthcare NLP Microservices: Developed and maintained Flask / FastAPI-based APIs containerized with Docker and orchestrated through Azure ML + CI/CD, improving information retrieval speed by 12%.
- Cross-functional Collaboration: Partnered with data scientists, DevOps, and clinical SMEs across global teams to translate healthcare data challenges into deployable, compliant ML solutions aligned with responsible AI practices.

Tech Stack Used: *Python, spaCy, TensorFlow, FastAPI, Azure ML, Databricks, CI/CD, SQL, PySpark, Azure Pipelines*

Software Developer Intern (Co-op) – Persistent Systems Limited, India | Dec 2021 – June 2022

- AI-powered Certification Verification System: Automated image & PDF validation pipeline using machine learning and OCR deployed via Azure ML pipelines, reducing manual verification by 70% and operational costs by 35%.
- Built NLP and machine learning models using spaCy, TensorFlow, and scikit-learn, improving prediction accuracy and data processing speed.
- Developed modular Python ETL and Feature engineering with SQL, Spark, and PySpark for large structured datasets (>200k rows). Deployed ML models to production environments using Azure ML, Databricks, Docker, and CI/CD tooling.

Tech Stack Used: *Python, TensorFlow, scikit-learn, spaCy, OCR, Azure ML, Databricks, Docker, SQL, Spark, PySpark*

EDUCATION

Master of System Science and Engineering (Interdisciplinary Artificial Intelligence) University of Ottawa, ON	2023 – 2025
Bachelor of Engineering in Electronics Engineering Shri Ramdeobaba College of Engineering and Management, India	2018 – 2022

AWARDS AND ACHIEVEMENTS

• Research Fellowship Scholarship, University of Ottawa, ON	2025
• Elected as Program Co-Ordinator for Systems Science & Eng. as a liaison between students and faculty.	2023
• Top Talent 2023, Persistent Systems (for measurable ML project impact), Persistent Systems Limited, India	2023
• 2nd Runner-Up, Global AI Hackathon Semicolons 2023, India	2023

PUBLICATIONS

• <u>Audio + Text based Emotion Recognition System</u>	2021
• <u>Detection of COVID-19 using Machine Learning and Android App Development</u>	2021

RELEVANT PROJECTS

Applied AI Research: LLM-Based Analytics & Predictive Modeling

- Designed and evaluated end-to-end AI/ML pipelines combining LLM-based reasoning (LangChain, Gemini, RAG) with supervised statistical models (LR, RF, GBDT), applying experiment design, hypothesis testing, feature engineering (SQL/Spark), and model selection across structured and unstructured datasets.
- Built and deployed production-ready components using FastAPI, Docker, and MLflow, enabling scalable inference, reproducible experimentation, and measurable performance gains ($\approx 45\%$ latency reduction, ROC/AUC ≈ 0.83).
- Integrated explainability and optimization techniques including SHAP, misclassification cost analysis, and sensitivity-specificity trade-offs to support transparent evaluation, bias detection, and business decision-making.